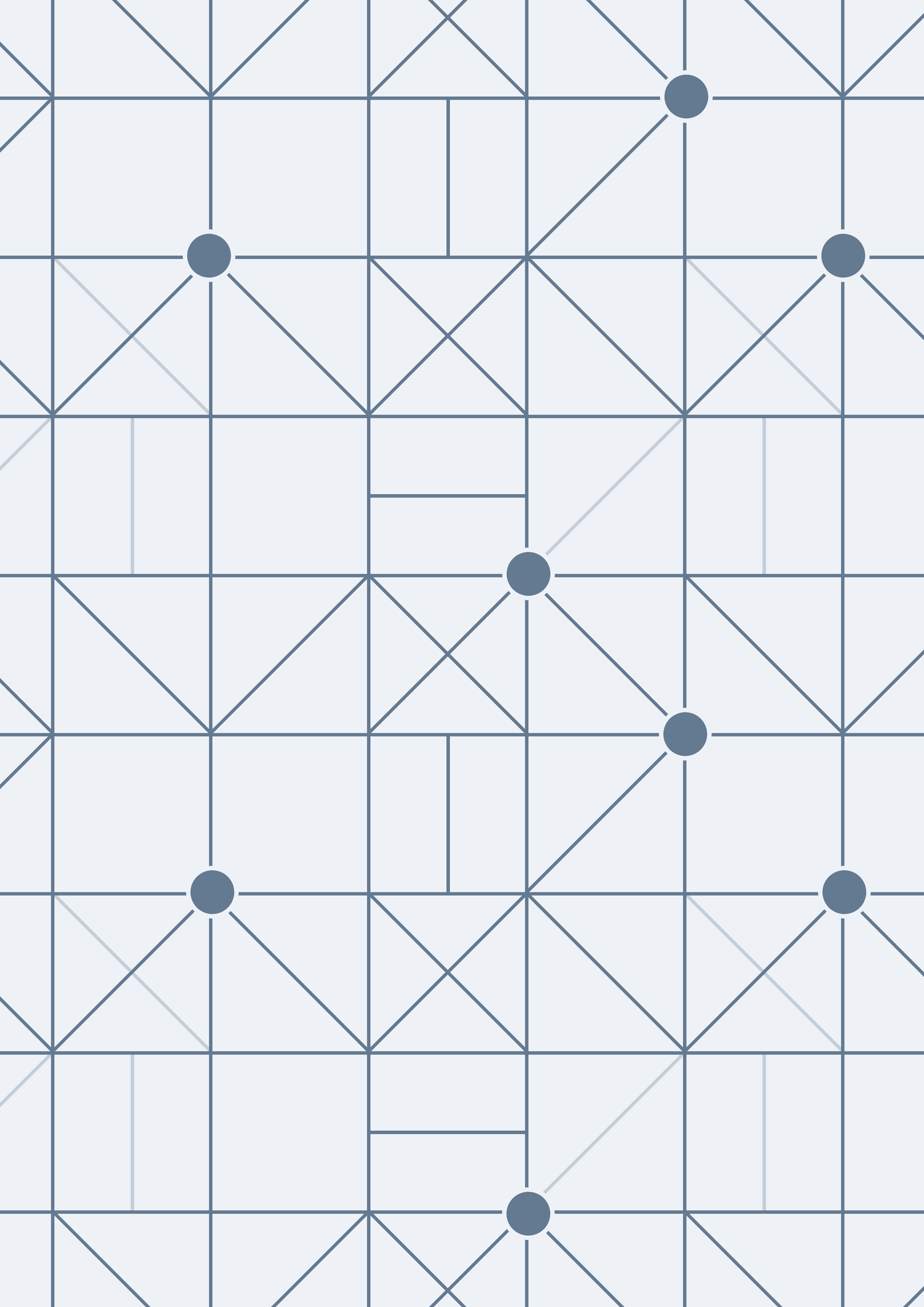


Microcanonical ensemble → E = cte ← but this is not satisfied in any real system, it's always dissipation

Then why do we study it? Because it's the mathematical frame to get the another ensembles.

POSTULATE OF AVERAGES

POSTULATE OF EQUAL A PRIORI PROBABILITY

The **POSTULATE OF AVERAGES** states that, for a system in thermal equilibrium, the time average of a physical quantity over a sufficiently long period is equal to the ensemble average for the same quantity in the corresponding ensemble.

The **POSTULATE OF EQUAL A PRIORI PROBABILITY** asserts that, in the absence of any additional information or constraints, all microstates consistent with a given macrostate are equally probable.

The two postulates represent the building blocks with which we will construct the ensembles of thermodynamic systems in equilibrium, starting with the simplest one: the ***isolated system****.*

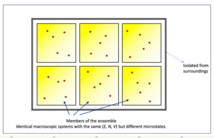
MICROCANONICAL ENSEMBLE: the members of the ensemble are macroscopic systems *identical copies of the original,* **ISOLATED** and **IN EQUILIBRIUM**.

ISOLATED SYSTEM: a system that does not interact thermally, mechanically, or materially with the surroundings; characterised by *adiabatic, rigid, and impermeable walls*.

$V = cte \qquad N = cte \qquad \boxed{H = E}$

IN EQUILIBRIUM: there is *no time dependence* of any variable, and the macrostate can be described using Thermodynamics. The density matrix/density function will also be independent of time.

Ensemble of isolated and equilibrium systems: macrostate (N, V, E)



MICROCANONICAL ENSEMBLE: the members of the ensemble are macroscopic systems *identical copies of the original,* **ISOLATED** and **IN EQUILIBRIUM**.

In the **classical** case, a microstate of the system is characterised by a point in phase space, Γ.

This phase space encompasses all possible combinations of positions and momenta for each particle in the system, and each point within this space represents a unique microstate:

$$(q,p) \equiv (q_1, p_1, \ldots, q_s, p_s) \longrightarrow 2s \text{ dimensions}$$

$\rho(\mathbf{q}, \mathbf{p}) \rightarrow$ probability density for the system to be in the microstate $(\mathbf{q}, \mathbf{p})$ (consistent with the macrostate)

What probability (statistical weight) do we assign to each of these microstates?

- **ISOLATED: a system that does not interact thermally, mechanically, or materially with the surroundings.**

$$V = \text{const} \quad N = \text{const}$$

$$H(\mathbf{q}, \mathbf{p}) = E = \text{const} \quad (q,p) \equiv (q_1, \ldots, q_{3N}, p_1, \ldots, p_{3N})$$

- **IN EQUILIBRIUM: the density function is stationary.**

Teorema de Liouville

$$\frac{\partial \rho}{\partial t} = 0 \Longrightarrow \boxed{\rho(q,p) = \rho\big(H(q,p)\big)}$$

WHAT IS THE DENSITY FUNCTION FOR THE MICROCANONICAL ENSEMBLE?

With the help of the postulate of *equal a priori probability*, it is possible to construct this density function clearly and directly.

The density function for the microcanonical ensemble is *constant* for all microstates consistent with the macrostate, assigning them equal probability.

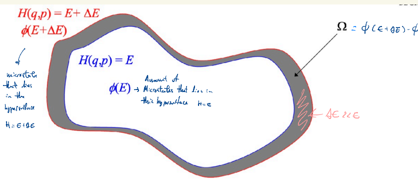
In other words, for a system in a microcanonical equilibrium, ***the density function is uniform in phase space****, reflecting the equal a priori probability of occupying any allowed microstate.*

An isolated system with energy *E* is an idealisation. ⟹

- In reality, we can never completely eliminate interactions with the external world.
- Any small error in determining the energy covers a very broad spectrum of microstates.
- According to the Heisenberg Uncertainty Principle, obtaining the exact energy would require a measurement of infinite duration.

Instead of specifying the energy, we will define an isolated system as one whose energies fall within the interval *ΔE*.

$$\boxed{\begin{gathered}[E, E+\Delta E] \\ \Delta E << E\end{gathered}}$$



$$\boxed{\Omega(E;\Delta E;N,V) \equiv \Omega(E) = \phi(E+\Delta E) - \phi(E) \approx \frac{\partial \phi}{\partial E}\Delta E = \omega(E)\Delta E}$$

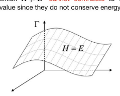
We'll see that ΔE is not relevant

↳ Taylor approximation

$\Omega(E, \Delta E, N, V) \simeq \frac{\partial \phi}{\partial E} \Delta E = \omega(E) \Delta E \Longrightarrow \omega(E) = \frac{\Omega}{\Delta E} \rightarrow$ number of microstates in ΔE

$\frac{\partial \phi}{\partial E}$: Density of states, ω ; $\omega(E)$: density

1. Any average over this ensemble must take into account that all mechanical states $(\mathbf{q}, \mathbf{p})$ for which $H \neq E$ cannot contribute to the mean value since they do not conserve energy.



2. ρ is a function exclusively of H.
3. Equal *a priori* probability for all microstates with $H = E$.
4. Normalisation condition: $\int \rho(\mathbf{q}, \mathbf{p})\, dq dp = 1$

Dirac delta excludes non-compatible microstates

$$\boxed{\begin{gathered}\rho(q,p) = \frac{1}{\omega(N,V,E)}\delta(E - H) \\ \omega(N,V,E) = \int \delta(E - H(q,p))\, dq\, dp\end{gathered}}$$

$\omega(N V E) = \frac{\partial \phi}{\partial E} = \int_\Gamma \delta(E-H)\, d\bar{q}\, d\bar{p} \Rightarrow$ Liouville's theorem $\Rightarrow \rho = cte$ where $H = E = cte$

$\rho(\bar{q}, \bar{p}) = \rho(H) = \frac{1}{\omega}\delta(E-H)$

$$\rho(\bar{q}, \bar{p}) = \begin{cases} 1/\omega & E \le H \le E+\Delta E \\ 0 & \text{elsewhere} \end{cases}$$

$$\rho(q,p) = \frac{1}{\omega(N,V,E)}\delta(E-H)$$

$$\omega(N,V,E) = \int \delta(E - H(q,p))\, dq\, dp$$

These expressions for the classical density function are not entirely correct

We need a correction factor to adjust dimensions and avoid issues with entropy

A. This gives a volume, not a number of states!!
   It needs correction → dividing by the "unit volume", h^s → we $\Delta p \Delta q \geq h$, the volume that occupy a single state then $\frac{\text{volume}}{\text{volume/particle}}$ = n° particle

B. If particles are identical, swapping two particles **does not give a new state.**
   Classical integral counts them as different → overcounting.
   Divide by $N!$ → correct number of states

J. G. Kirkwood Phys. Rev., **44** (193
J. G. Kirkwood, Phys. Rev., **45** (19

↳ without this coefficient, the entropy in the real world would be a intensive variable. this mean that larger N, the entropy do not increase } Revision

Final expressions

$$\rho(q,p) = \frac{1}{h^s N!}\frac{1}{\omega(N,V,E)}\delta(E-H)$$

$$\omega(N,V,E) = \frac{1}{h^s N!}\int \delta(E - H(q,p))\, dq\, dp$$

Mean value of the dynamic function $a(\mathbf{q},\mathbf{p})$

$$\langle a(q,p)\rangle = \frac{1}{h^s N!}\frac{1}{\omega(N,V,E)}\int a(q,p)\delta(E - H(q,p))\, dq\, dp$$

$$\langle a(q,p)\rangle = \int a(q,p)\rho(q,p)\, dq\, dp = \frac{1}{h^s N!}\frac{1}{\omega(N,V,E)}\int a(q,p)\delta(E - H(q,p))\, dq\, dp$$

Ω is the most important quantity in the microcanonical ensemble.

From Ω, and its dependence on the variables N, V, and E, the **complete thermodynamics** of the system can be deduced.

What is the connection between Ω and thermodynamics?

Lets see that without $\frac{1}{h^s N!}$ $\int_\Gamma \rho(q,p)\, dq\, dp \neq 1$

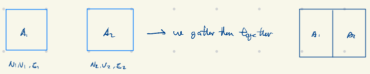
$$1 = \int_\Gamma \rho(q,p)\, dq\, dp = \int_\Gamma \frac{1}{\omega}\delta(E-H)\, dq\, dp = \frac{1}{\omega}\int_\Gamma \delta(E-H)\, dq\, dp = \frac{h^s N!}{\int_\Gamma \delta(E-H)\, dq\, dp} \quad \int_\Gamma \delta(E-H)\, dq\, dp = h^s N! \neq 1$$

$$\langle a\rangle = \int_\Gamma a(q,p)\rho(q,p)\, dq\, dp = \int_\Gamma a(q,p)\frac{1}{h^s N!}\frac{1}{\omega}\delta(E-H)\, dq\, dp = \frac{1}{h^s N! \omega}\int_\Gamma a(q,p)\delta(E-H)\, dq\, dp$$

$\boxed{\ln \Omega}$ extensive function → $\ln\Omega = \ln\Omega_1 + \ln\Omega_2$

this what we wanna show

Isolated systems



$E = E_1 + E_2 + E_{12} \simeq E_1 + E_2 = cte$

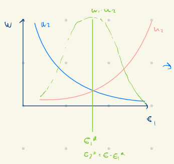
The Hamiltonian of the whole system $A_1 + A_2$

$H = H_1 + H_2$

$$\omega = \frac{1}{h^s N!}\int_\Gamma \delta(E-H)\, dq\, dp = \frac{1}{h^{s_1+s_2}N_1!N_2!}\int_\Gamma \delta(E-H_1-H_2)\, dq_1 dp_1 dq_2 dp_2 = \frac{1}{h^{s_1}N_1!}\int_\Gamma dq_1 dp_1 \frac{1}{h^{s_2}N_2!}\int_\Gamma \delta(E-H_1-H_2)\, dq_2 dp_2$$

↳ $\times \int_0^E \delta(E_1 - H_1)\, dE_1 = 1$ $\quad 0 \leq H_1 \leq E$

$$\int_0^E dE_1 \underbrace{\frac{1}{h^{s_1}N_1!}\int_\Gamma \delta(E_1-H_1)\, dq_1 dp_1}_{\omega_1(E_1)} \underbrace{\frac{1}{h^{s_2}N_2!}\int_\Gamma \delta(\overbrace{E-E_1}^{E_2}-H_2)\, dq_2 dp_2}_{\omega_2(E-E_1)} = \int_0^E \omega_1(E_1)\omega_2(E-E_1)\, dE_1$$



→ We assume that
$$\int_0^E \omega_1(E_1)\omega_2(E-E_1)\, dE_1 \simeq \int_0^E \omega_1(E_1^*)\omega_2(E-E_1^*)\, e^{-\frac{(E_1-E_1^*)}{2\sigma_E^2}}\, dE_1 = \omega_1(E_1^*)\omega_2(E-E_1^*)\int_{-\infty}^{+\infty} e^{-\frac{(E_1-E_1^*)}{2\sigma_E^2}}\, dE_1 \Rightarrow \omega(E) = \omega_1(E_1^*)\omega_2(E-E_1^*)\sqrt{2\pi}\,\sigma_E$$

Gaussian distribution integral

ω_1, ω_2 are constants
$\omega_1(E_1^*)$ $\omega_2(E-E_1^*)$

I extended the limits, the most contribution to the integral are the points near to E_1^*, the far points' contributions are negligible → $\int_{-\infty}^{\infty} \sim \int_0^E$

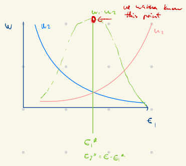
$\omega(E) = \omega_1(E_1^*)\omega_2(E-E_1^*)\sqrt{2\pi}\,\sigma_E$

Let's remember the definition

$\omega = \frac{\Omega}{\Delta E}$

} $\times \Delta E$ → $\omega\Delta E = \omega_1\Delta E\, \omega_2\Delta E \frac{\sqrt{2\pi}\,\sigma_E}{\Delta E} = \Omega = \Omega_1\Omega_2\frac{\sqrt{2\pi}\,\sigma_E}{\Delta E}$ → Take logarithm $\ln\Omega = \ln\Omega_1 + \ln\Omega_2 + \ln\frac{\sqrt{2\pi}\,\sigma_E}{\Delta E} \simeq \ln\Omega_1 + \ln\Omega_2$ → we showed that the $\ln\Omega$ is an extensive value from $E \simeq E_1 + E_2$

This value very slowly



We seek the equilibrium ← Next revision

$$\frac{d}{dE_1}\left(\omega_1(E_1)\omega_2(E-E_1)\right) = 0 \Leftrightarrow \frac{d\omega_1(E_1)}{dE_1}\bigg|_{E_1^*}\omega_2(E-E_1^*) + \omega_1(E_1)\frac{d\omega_2(E-E_1)}{dE_1}\bigg|_{E_1^*} = 0 \Leftrightarrow \frac{1}{\omega_1}\frac{d\omega_1}{dE_1}\bigg|_{E_1^*} - \frac{1}{\omega_2}\frac{d\omega_2}{dE_2}\bigg|_{E_2^*} = 0 \Leftrightarrow \frac{1}{\Omega_1}\frac{d\Omega_1}{dE_1}\bigg|_{E_1^*} - \frac{1}{\Omega_2}\frac{d\Omega_2}{dE_2}\bigg|_{E_2^*} = 0 \rightarrow \overbrace{\frac{d\ln\Omega_1}{dE}\bigg|_{E_1^*}}^{\beta_1} = \overbrace{\frac{d\ln\Omega_2}{dE_2}\bigg|_{E_2^*}}^{\beta_2}$$

$E_1 + E_2 = cte \rightarrow dE_1 = -dE_2$
↳ $-\frac{d\omega_2(E_2)}{dE_2}\bigg|_{E_2^*}$

there's two properties that are identical when we reach the equilibrium